New paleomagnetic results from the Neoproterozoic successions in southern North China Block and paleogeographic implications

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Abstract A total of 312 oriented core samples collected from the Neoproterozoic to lower Cambrian strata in the Lushan region of western Henan Province, southern North China Block, were subjected to stepwise demagnetization. After removing recent and possibly Mesozoic overprint components, inferred primary magnetic remanence was isolated from the Precambrian Dongjia, Sanjiaotang and Cuizhuang Formations. Our new results, together with selected previous results, suggest that the North China Block was in low latitudinal positions for most of the studied time interval. Through comparison with paleogeographic positions of other continents, we suggest that the NCB could have been closed to Siberia during the ca. 800–600 Ma interval.

Keywords: North China Block, Neoproterozoic, paleomagnetism, paleogeography.

Paleomagnetic investigations have previously been carried out in the North China Block (NCB) in order to understand its paleogeographic positions. These include the study of the Proterozoic type section in Jixian[1—5], the late Precambrian succession in eastern Liaoning Province[6, 7], and the late Precambrian–Cambrian succession in the Xu-Huai region[8, 9]. However, severe conflicts exist among these published results for rocks of similar ages, sometimes from the same geological section. Possible reasons for the conflicts include: (1) magnetic overprints not effectively removed, (2) insufficient sample numbers studied in some work, and (3) modern demagnetization and data analysis techniques not properly followed.

In this paper, we report new Neoproterozoic and Cambrian paleomagnetic results from the south margin of the NCB. Based on those results and a critical selection of existing data, we discuss possible paleogeographic positions of the NCB in the Neoproterozoic supercontinent reconstructions.

1 Geological framework of the study region

The Proterozoic cover succession in the NCB has been divided into the Changchengian (ca. 1 800—1 400 Ma), the Jixianian (ca. 1 400—1 000 Ma), the Qingbaikouan (ca. 1 000—800 Ma)

and the Sinian (ca. 800—545 Ma) systems. The type section of the first three systems is in Jixian. There is a sedimentation hiatus through the Sinian to the earliest Cambrian in Jixian, but strata of that age interval are well developed in Fuzhou, Xu-Huai and along the southern margin of the craton. We have selected the Lushan region in western Henan Province for this study (fig.1) because (1) there is a fairly complete succession from the late Mesoproterozoic to the Lower Cambrian, and (2) the structure there is simple.

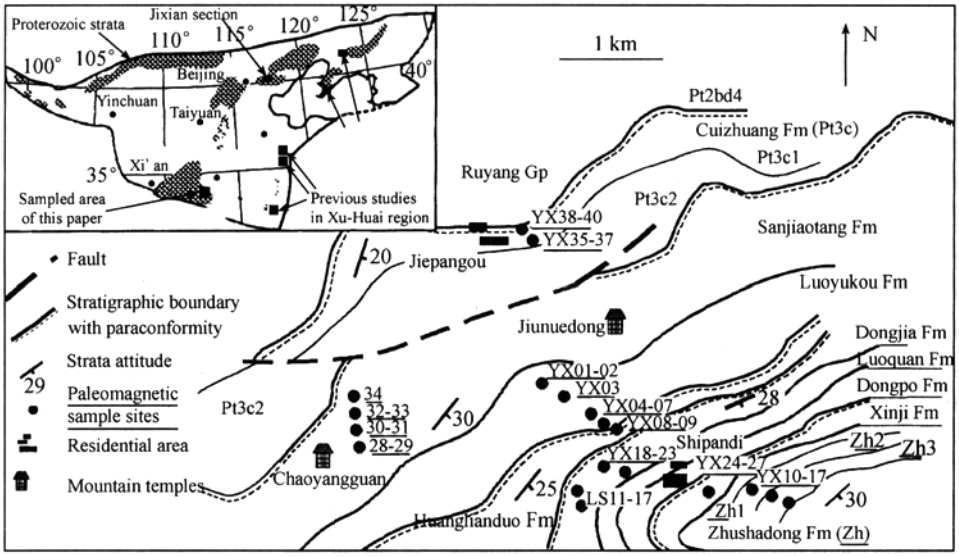


Fig. 1. Tectonic position and the structural map of the sampled area (112.7°E; 33.8°N).

In the study region, the Changchengian is represented by the Xionger Group, composed predominantly of volcanic rocks with isotopic ages between 1.85 and 1.40 Ga[10]. Discomformably overlying the Xionger Group is a ~2 000 m thick well exposed continental shelf succession. According to lithostratigraphic and biostratigaphic analysis, this succession is of late Jixianian to the Early Cambrian age[11—15]. In this study we collected paleomagnetic samples from strata ranging from Neoproterozoic to the Early Cambrian (fig. 2) in the well-known Jiepangou-Jiunuedong-Shipandi section[15] (fig. 1). The strata in general bear a gentle dip toward the southeast with dip angles ranging between 15° and 35°. There is no widespread penetrative deformation in the rocks.

Although the chronological control for the strata is largely established through biostratigraphic and regional lithostratigraphic correlation within the NCB, reliable absolute ages are rare. Available age data[12—14] in the study region are listed in fig. 2. Some of the ages are apparently out of stratigraphic order, and should be treated with extreme caution. Sandstones in different formations bear abundant glauconite that yielded late Precambrian ages, indicating that the region has not undergone any severe thermal events since the authogenic minerals were formed.

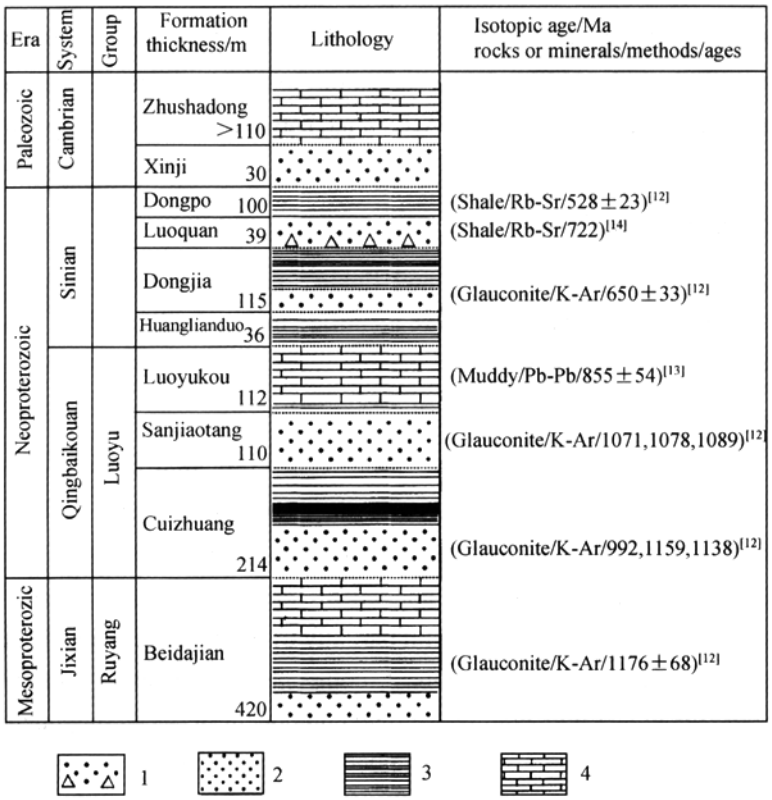


Fig. 2. Sampled succession and isotopic data in Lushan region. 1, Tillite; 2, sandstone; 3, mudstone & shale; 4, carbonatite.

2 Sampling and measurements

Three hundred and twelve core samples were collected using a gasoline-powered drill during the field trips of 1997 and 1999. Samples were oriented using both a magnetic compass and a sun compass when possible. Each core sample was cut into 1—3 ca. 22 mm-long specimens. The majority of experiments were carried out in the Paleomagnetic Laboratory in the University of Western Australia (UWA), and 67 specimens were demagnetized in the Paleomagnetic Laboratory in the Geological and Geophysical Institute of the Chinese Academy of Sciences (CAS). Remanence measurements in both laboratories were carried out using 2G-Enterprise three-axis cryogenic magnetometers housed in room size Helmholtz coils, with a low magnetic field (＜100 nT) in the sample-loading region. AF demagnetization was carried out using a 2G600 automated AF demagnetizer attached to the cryogenic magnetometer, and thermal demagnetization using MMTD1 (in UWA) and MMTD60 (in CAS) furnaces, also hosted within the Helmholtz coil system. During the experimental procedure, samples were further protected from magnetic contamination using Mumetal boxes. All samples were stepwise demagnetized. Thermal treatments were applied to all the reddish colored specimens. The temperature steps range from as large as 150℃ at the

low-temperatures to as small as 5℃ at around 570℃ and above 600℃. Some samples were treated by low-temperature (＜300℃) demagnetization first to eliminate any remanence carried by goethite, and followed by alternating field (AF) demagnetization to isolate any remanence carried by magnetite.

3 Magnetic component analysis

Both a CSIRO-ROCKMAG package and Enkin's paleomagnetic software (1994 version) were used for data analysis. Components were identified mainly by line fitting[16] and determining stable endpoints. The results are summarized in table 1, and are discussed below.

3.1 The Cambrian samples

The gray limestone samples from the Zhushadong Formation (sites YX10—YX13) have natural remanent magnetization (NRM) intensities in the order 1 mAm^{-1} and in most cases are reduced to half of their NRM values by thermal demagnetization below 250℃. A combined thermal and AF demagnetization method was applied to some weakly magnetized samples. Many samples exhibit two magnetic components with partially overlapping demagnetization spectra (fig. 3 (a)). The low temperature/AF field component (component A hereafter) is oriented to the north and downward, and the higher temperature/AF field component (termed component B hereafter) is oriented to the northwest and downward. Component A is sub-parallel to the present local geomagnetic field direction and was found in the majority of the samples throughout this study.

The drab to purple muddy carbonate samples from the Zhushadong Formation (sites YX14—YX17) have NRM intensities in the range of 1—20 mAm^{-1}. The majority of these samples have unblocking temperatures over 575℃. In most cases they yield component A before 350℃ and component B between 400—600℃. Fifteen percent of the samples exhibit stable endpoints during the 550—635℃ steps, which is oriented *in situ* shallow to the northwest. We term it the component C.

NRM intensities of the gray sandstone samples from the Xinji Formation (sites YX24—YX27) are in the range of 0.5—10 mAm^{-1}. The majority of the samples yielded component A and, in about one third of the samples, it is the only clearly defined component. A little more than 10 percent of the samples yielded a high temperature component that is oriented downward to the southeast (fig. 3 (b)). After tilt correction, this direction overlaps with the antiparallel direction of the bedding-corrected component C from the Zhushadong Formation (table 1).

3.2 The Sinian samples

Samples from the Dongjia Formation are mainly purple muddy dolomite (sites YX18—YX23, and LS11—LS17). They have NRM intensities of about 1—10 mAm^{-1}. Results obtained

Table 1 Paleomagnetic results from the Neoproterozoic and Cambrian strata of the Lushan region, western Henan Province

Demagnetization	Comp.	*N*	*Nr*	Coor.	*D*	*I*	*k*	α_{95}	Plon	Plat	Dm	Dp	P
Zhushadong Fm (8 sites, 54 samples)													
NRM-350 °C/30mT	A	47	0	G	349.7	45	22.5	4.5	346.2	78.5	5.7	3.6	
				S	28.6	57.1	23.6	4.4	184.8	66.6	6.4	4.7	
400—600℃	B	44	11	G	327.2	44.1	10.3	7.0	15.8	60.6	8.8	5.5	PB
				S	2.9	68.4	9.3	7.5	118.5	72.1	12.6	10.7	
550—635℃	C	7	1	G	341.5	–33.2	13.0	17.4	314.3	35.2	19.7	11.2	
				S	333	–11.6	13.3	17.2	330.7	42.8	17.5	8.9	PC1
Xinji Fm (6 sites, 24 samples)													
NRM-460℃/30mT	A	21	0	G	4.6	50.3	13.5	9.0	236.6	85.2	12.1	8.1	
				S	48.3	67.2	13.0	9.2	163.0	51.4	15.3	12.7	
430—595℃/115mT	A	10	2	G	3.0	39.2	11.9	14.6	279.1	78.1	17.4	10.4	
				S	28.5	60.1	11.8	14.7	176.6	66.4	22.3	16.8	
430—595℃/115mT	C	4	3	G	336.7	–46.1	22.3	19.9	315.4	24.9	25.5	16.3	
				S	334.2	–16.3	22.1	20.0	327.6	41.3	20.6	10.6	PC2
Mean for Cambrian	C	11	4	G	339	–38	15.1	12.2	315.6	31.3	14.4	8.5	PC′
				S	333.5	–13.3	16.8	11.5	329.4	42.3	11.7	6.0	PC
Dongjia Fm (13 sites, 92 samples)													
NRM-490℃	A	39	0	G	355.6	47.2	97.7	2.3	328.6	83.4	3.0	1.9	
				S	32.0	63.8	93.6	2.4	167.5	63.0	3.8	3.0	
490—645℃	A	78	0	G	7.1	54.8	23.4	3.4	186.0	84.0	4.8	3.4	
				S	60.7	74.3	19.0	3.8	148.5	43.2	6.9	6.2	
490—645℃	D	26	20	G	6.9	–13.0	10.3	9.3	282.1	49.1	9.5	4.8	DJ′
				S	7.4	10.7	12.2	8.4	277.4	60.8	8.5	4.3	DJ
Huanglianduo Fm (2 sites, 12 samples)													
NRM-600℃	A	10	1	G	342.7	59.9	45.4	7.2	54.6	74.6	10.9	8.2	
				S	50.3	80.1	27.3	9.4	133.5	44.4	18	17.3	
Luoyukou Fm (7 sites, 42 samples)													
NRM-490℃	A	39	0	G	355.7	53.7	17.5	5.6	30.9	86.4	7.8	5.5	
				S	41.5	70.7	17.0	5.7	153.4	54.4	9.9	8.6	
450—600℃	B1	42	0	G	331.9	54.8	14.9	5.9	34.4	66.9	8.4	5.9	PB1
				S	359.1	80.9	13.7	6.2	112.2	51.6	12.0	11.5	
Sanjiaotang Fm (8 sites, 51 samples)													
NRM-670℃	A	26	2	G	353.8	52.1	9.5	9.7	12.5	84.7	13.3	9.1	
				S	41.4	61.2	10.2	9.3	175.5	56.7	14.3	11.0	
500—650℃	E	20	14	G	42.8	14.4	10.6	10.5	226.6	42.5	10.8	5.5	SJT′
				S	49.0	8.8	11.6	10.0	224.4	35.9	10.1	5.1	SJT
Cuizhuang Fm (6 sites, 37 samples)													
NRM-400℃	A	22	0	G	359.5	45.9	8.0	11.7	296.6	83.5	14.9	9.5	
				S	11.7	49.1	6.0	13.9	220.4	79.4	18.4	12.2	
550—660℃	F	19	0	G	33.3	24.2	52.0	4.7	229.5	53.1	5.0	2.7	CZ′
				S	43.3	27.8	65.0	4.2	217.8	46.6	4.6	2.5	CZ
480—670℃	AR	10	10	G	13.2	38.3	17.9	11.8	245.7	73.1	13.9	8.2	
				S	13.9	25.1	16.1	12.4	257.8	65.8	13.3	7.2	

Comp., component name; *N*, number of samples used for statistical average; *Nr*, number of the reversed polarity samples; Coor., G/S, geographic/stratigraphic coordinates; *D*, declination; *I*, inclination; *k*, Fisher precision parameter for direction; α_{95}, radius of 95% confidence circle of the direction. Plon/Plat, east longitude/north latitude of VGP; Dm/Dp, 95% error of the VGP. P, abbreviated pole names.

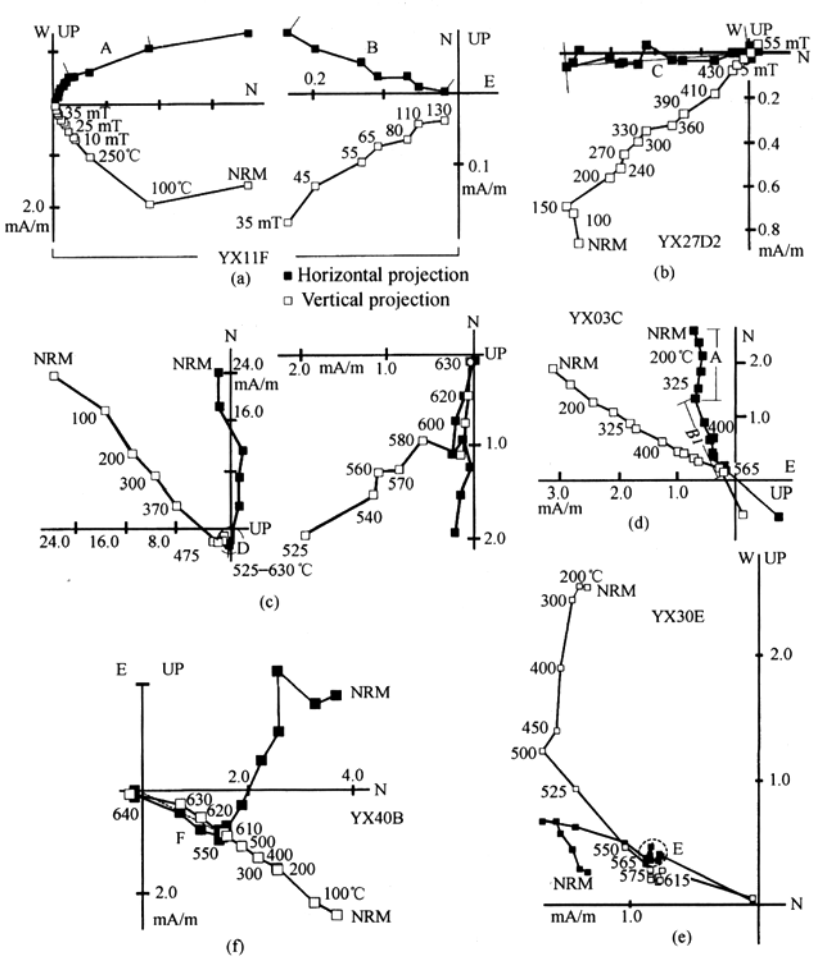


Fig. 3. Examples of orthogonal plots showing identified major components (all results are *in situ*).

independently from the two laboratories agree with each other. The demagnetization paths can be classified into two types: one is characterized by a single component A, and the other by well-defined great circles between component A and a component defined by high-temperature stable endpoints or vectors. This high-temperature component is oriented to the south with very shallow inclinations, which is different from the previously defined components, and is thus regarded as a new component (D) (fig. 3(c)). Some samples, although exhibiting similar great circles, failed to reach a stable endpoint at high temperature, and therefore could not be used for defining component D. Six samples yielded remanent directions shallow to the north, making component D a dual-polarity one.

Samples of the Huanglianduo Formation (sites YX08—YX09) are from purple dolomite. Their NRM intensities are in the order of 0.1 mAm^{-1}. Only component A has been isolated before the remanence intensities increase at the temperature range of 550—575℃, accompanied by randomization of remanence directions.

3.3 The Qingbaikouan samples

Samples of the Luoyukou Formation are from purple dolomite (sites YX01—YX09). Their NRM intensities are mostly in the order of 1 mAm^{-1}. In most cases, two components could be clearly isolated: component A before 490℃, and a poorly-resolved component oriented northwesterly downward between 450—600℃(fig. 3(d)), similar to component B from the Zhushadong limestone samples. This component is termed B1.

Samples from the Sanjiaotang Formation (sites YX28—YX34 and LS25) are purple sandstones, with NRM intensities in the order of 1 mAm^{-1}. Slightly over half of the samples are dominated by component A, and the rest of the samples revealed southwesterly upward, or northeasterly downward, directions defined by stable endpoints in the temperature range of 500—650℃(fig.3 (e)). We term this dual-polarity component component E. Low-temperature components that coexist with component E are randomly directed.

Samples from the Cuizhuang Formation include purple sandstone samples from the lower part of the formation (sites YX38—YX40), and red muddy sandstone samples from the oolitic hematite layer (sites YX35—YX37). The demagnetization paths of the sandstone samples are characterized by a high-temperature (ca. 550—660℃) component oriented downward toward the northeast (fig. 3 (f)). This direction is statistically different from component E with 95% confidence and is thus termed component F. The removed soft component from 14 out of the 19 samples has a mean direction close to that of component A. The oolitic hematite samples normally have stronger NRM intensities (ca. 10 mAm^{-1}). One third of these samples were dominated by component A, and the remainders yielded a vector antiparallel with that of component A. We term it component AR. It is possible that all the hematite samples were remagnetized due to Quarternary weathering.

3.4 Ages of the remanence components

As discussed above, component A is clearly a modern field overprint (table 1). Reversed directions of component A are present in both the Xinji and Huanglianduo carbonate rocks, in the Sanjiaotang purple sandstones, and in the hematite bed of the Cuizhuang Formation, possibly reflecting prolonged weathering process. This component is present in the entire succession, and in some case, is the only remanence component with unblocking temperature up to ca. 650℃.

Component B is less common than A, but it nonetheless appeared in more than one level of the stratigraphic succession, particularly in carbonate rocks. Paleopoles calculated according to the *in-situ* directions of component B overlap the Triassic paleopoles of NCB[17] (poles PB′ and PB1′ in fig. 4 (a)), indicating that it is likely a Mesozoic overprint related to the collision between the North and South China Blocks[18]. Its high unblocking temperature (up to 600℃) suggests a chemical origin, possibly related to orogenic fluids. However, the rare occurrence of such overprint in the majority of the samples suggests that the Mesozoic overprint event is not pervasive in

the study region.

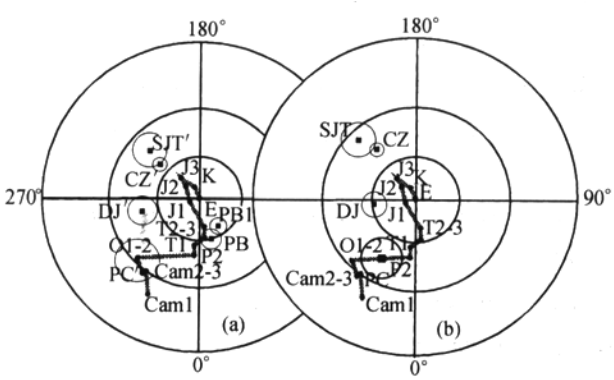


Fig. 4. (a) Poles PC′, DJ′, SJT′ and CZ′ (solid squares) based on the main components (*in situ*) with 95% confidence circles, plotted against existing Phanerozoic poles (solid dots) and APWP for the NCB[17]; (b) poles PC, DJ, SJT, CZ (solid squares) calculated from components C, D, E, F (tilt corrected) respectively. Equal area projections.

Paleopole calculated from *in situ* components C falls closer to the paleopoles of Middle to Late Cambrian and Ordovician ages rather than the Early Cambrian one from the NCB[19, 20] (fig. 4 (a)), and the difference becomes slightly greater after bedding-correction (fig. 4 (b)). Considering that there was no Cambro-Ordovician tectonic event interior North China that might have caused a remagnetisation, we interpret the components C as primary remanence and the difference between the PC pole and the Cam1 pole as reflecting the apparent polar wandering.

Components D, E, F are all interpreted as primary remanence for the following reasons. (1) They are different from each other despite of the common rock types between some stratigraphic units. (2) The calculated paleopoles are in general different from the expected Phanerozoic poles of the NCB [17] (fig. 4). (3) The studied stratigraphic succession was only mildly deformed with no sign of metamorphism or thermal event since the deposition of the rock formations. (4) Components D and E both possess dual-polarities.

4 Paleogeographic positions of the NCB

4.1 Quality assessment of the available paleomagnetic database

In order to interpret the new results presented above, a thorough review of the published data from the Neoproterozoic strata in the NCB is necessary. Previous workers have already noticed the problem of remagnetizations[3] and conflicting results[21] among some of those data. In this study, we applied the 7-point criteria[22] (Van der Voo, 1990) to select the more reliable data. We rejected a pole only when there is enough evidence to suggest that it is an overprint pole. We retained some data of questionable quality which, although not used for our reconstruction, may still be proved to be primary by future studies. The poles selected for paleogeographic reconstruction are listed in table 2.

Age uncertainty has long been the most serious problem for Precambrian paleomagnetism. Some researchers proposed that a pole should not be used for defining an APWP unless it has an age uncertainty of <30 Ma[23]. Buchan et al. (2000)[24] required a "key pole" to have an age uncertainty of <20 Ma, and that the rock units should be dated generally by the U-Pb or ^{40}Ar-^{39}Ar techniques. According to such a criterion, there is no "key pole" in the Neoproterozoic of the NCB

at this stage because of their age uncertainty alone.

Table 2 Selected Cambrian to Neoproterozoic paleomagnetic poles from the NCB

No.	Abbr.	Rock unit/Region/Reference	*N*	α_{95} (A95)	Plat (° N)	Plon (° E)	Criteria 1	2	3	4	5	6	7	*Q*
1	CAM3	upper Cambrian /NCB/[20]	66	5.4	31.7	329.6	+	+	+	+	+	+	+	7
2	CAM2	middle Cambrian/NCB/[20]	86	5.5	37	326.7	+	+	+	+	+	+	+	7
3	CAM1	lower Cambrian/NCB/[20]	32	6.5	18.5	342.0	+	+	+	+	+		+	6
4	DJ	Dongjia Fm/Lushan / (this study)	26	8.4	60.8	277.4	+	+	+		+	+	+	6
5	ZQ	Zhangqu Fm/Xu-Huai/[8]	11	8.3	46.4	288.7	+		atp		+		+	3
6	NY	Niyuan Fm/Xu-Huai/[8]	6	8.8	44.0	294.3	+		atp		+		+	3
7	ZW	Zhaowei Fm/Xu-Huai/[8]	10	23.0	41.2	281.7	+		atp		+		+	3
8	JY	Jiayuan Fm */Xu-Huai/[8]	8	20.0	39.7	284.0	+		atp		+		+	3
9	A1	average poles ZQ, NY, ZW, JY	–	5.7	42.9	287.0								
10	NF	Nanfen Fm/Fuzhou/[7]	57	9.5	16.5	301.1	+	+	+		+		+	5
11	SJT	Sanjiaotang Fm /Lushan/this study	15	10	41.3	227.0	+		+		+	+	+	5
12	CZ	Cuizhuang Fm, Lushan / this study	19	4.2	46.6	217.8	+		+		+		+	4

N /(*n*), number of sites/samples; α_{95}/ (A95), radius of 95% confidence circle for a direction/pole. Plon/Plat, east longitude/north latitude of VGP; the 7 criteria *Q* value of Van der Voo (1990)[22]. Atp, partly demagnetization. * Jiayuan Fm now is combined into lower member of Jiuliqiao Fm (Xing et al, 1989)[12].

Isotopic age of the Cuizhuang and Sanjiaotang Formations are in the range of 992—1159 Ma and most of those are over 1000 Ma. However, many researchers like to correlate them with the Qingbaikouan Xiamaling Formation in the Jixian type section[12], which has a glauconite K-Ar age of 956 Ma[12] and a limestone Pb-Pb age of 897±18 Ma[13]. Together with a Pb-Pb age of 855±54 Ma from the Luoyukou Formation[13], we preliminarily assigned to the Cuizhuang and Sanjiaotang Formations an age range of ~900—950 Ma. In the Fuzhou region, the Nanfen Formation is correlated with the uppermost part of the Qingbaikouan, but it has a glauconite K-Ar age of only 787 Ma. Without any other evidence, we asigned an age of ~780—800 Ma for this formation. Poles 5, 6, 7 and 8, from the Zhangqu, Niyuan, Zhaowei and Jiayuan Formations respectively, agree well with each other. We thus combine them into an average pole A1, and assigned it an age of ~700 Ma according to the glauconite K-Ar ages for the Zhaowei (738 Ma)[8] and the Shijia (681 Ma)[8, 12] Formations. Pole DJ is assigned an age of ~650 Ma based on the glauconite K-Ar age of the Dongjia Formation, with an uncertainty of no less than 30 Ma.

Polarity determination for Precambrian paleomagnetic results is also a problem. The rule normally followed is to maintain a minimal distance between two poles that have the smallest age gap between them. In the following, we will start with the Cambrian poles for convenience.

Even the Cambrian poles from the NCB have uncertain polarities[19, 20]. The majority of the workers interpreted the primary components oriented to the northwest and upward as the normal polarity (polarity option 1 hereafter). However, based on the understanding that the earth magnetic field had a strong bias toward the reversed polarity in the Early Paleozoic, Zhao et al.[19] proposed an opposite polarity for the above-mentioned direction (polarity option 2 hereafter).

4.2 Paleogeography positions of the NCB

Fig. 5 shows the paleolatitudes and orientations of the NCB implied by the selected paleopoles according to polarity option 1. If one adopts the polarity option 2, the positions of the NCB would be the mirror image of the given position on the opposite side of the equator.

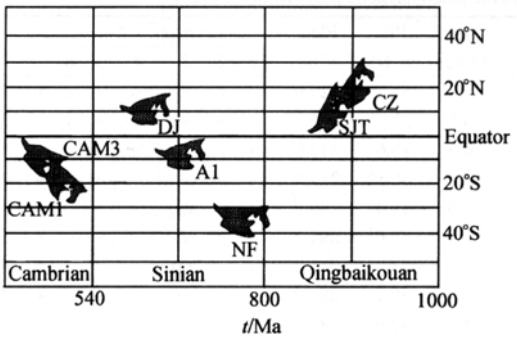


Fig. 5. Paleo-latitudes and orientations of the NCB implied by the selected poles in table 2.

In fig. 6, we plotted the possible positions of the NCB against other continents. In the ca. 780—800 Ma Rodinia reconstruction in fig. 6 (a), the position of Laurentia is controlled by the 780 Ma "key pole" selected by Buchan et al.[24]. East Gondwana-South China-Laurentia connection is after Li et al.[25]. Relation between Siberia and Laurentia is based on Hoffman [26]. For Baltica and other blocks, except NCB, the rotation parameters relative to Laurentia are from Torsvik et al.[27]. We take the pole from the Nanfen Formation (pole NF in table 2) for the ~780—800Ma reconstruction of the NCB. In the

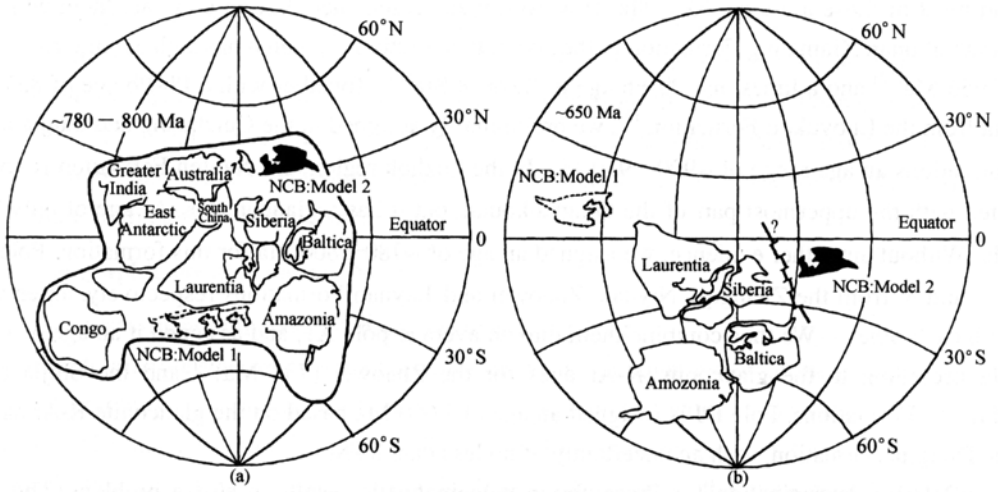


Fig. 6. Paleogeographic positions of the North China Block. (a) Rodinia at ~780—800 Ma. Model 1 position for the NCB is based on polarity option 1 of pole NF in table 2, and Model 2 based on polarity option 2. Connection of East Gondwana-SCB-Laurentia is based on Li et al.[25]. Euler pole for rotating Siberia relative to Laurentia is (11°N, 12°E, 31°counterclockwise). Other continents were rotated to Laurentia according to Torsvik et al.[27]. Position of the Laurentian pole is (9°N, 136°E)[24]. (b) ~650 Ma reconstruction. Model 1 position is based on polarity option 1 of pole DJ in table 2, and Model 2 based on polarity option 2. Euler pole for rotating Siberia to Laurentia is (11°N, 12°E, 31°counterclockwise). Other continents were rotated to Laurentia according to Torsvik et al.[27].

normal polarity (option 1) reconstruction, the NCB could be placed against the Greenville margin of Laurentia, or other positions at the same latitude (such as east of Amazonia). However, considering that there is no evidence suggesting a Grenville-age orogenic belt along the northern margin of the NCB, a position opposite to the Grenville Belt is regarded as unlikely. We thus prefer to

use the polarity option 2 in the reconstructions, and place the NCB against the Lake Baikal margin of Siberia in a northern latitudinal position.

During the breakup of Rodinia, Laurentia rotated largely clockwise[27]. Model 1 (according to polarity option 1 of pole DJ, table 3) of NCB's position would be along the tropic belt of north hemisphere, and model 2 position according to polarity option 2 of pole DJ still allows it to be next to Siberia (fig. 6 (b)). Given that the relative orientations between Laurentia-Siberia and the NCB had slightly changed from the 780—800 Ma reconstruction, the breakup between them may have already occurred by then. Qiao et al.[28] reported stratigraphic record of paleo-earthquakes, and dykes of ~720—600 Ma age, in eastern NCB. These activities could be related to the breakup. However, more reliable data from both continents are required to further evaluate the timing of such breakup.

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